

DeckForge AI — speaker notes

Bachelor defence · Kylian Algoet · 2026-09-02

Deck	26 slides (22 core, 4 supporting)
Estimated speaking time	26.4 min at 130 wpm, slides only — an estimate from the word count of these notes, not a measured rehearsal
Live demo	4 min target, timed separately — <code>docs/presentation/demo-script.md</code>
Backup plan	<code>docs/presentation/demo-backup-plan.md</code> — read it before presenting
Jury questions	<code>docs/presentation/jury-questions.md</code>
Built from	commit 685664b on 2026-08-14

No authoritative presentation duration is recorded anywhere in this repository. The slide count is provisional. If a duration is set by the school, cut the **supporting** slides first — each one folds into the note of the slide beside it, so a shorter talk loses stage time and not a claim.

CORE · title · ~55 s spoken (119 words)

ON THE SLIDE

Kylian Algoet · Multimedia & Creative Technologies
Erasmushogeschool Brussel · Bachelor defence, 2 September 2026

SAY

Good morning. I am Kylian Algoet, and this is DeckForge AI.

It lets someone describe a skateboard deck graphic in words, pick a style, optionally give it a reference image, and see the artwork on a 3D deck before downloading it. The model is fine-tuned locally, on this laptop, on a dataset I built.

What I am defending is not the demo. The demo takes four minutes and it works. It is the research: a measured account of what fits in eight gigabytes of video memory, what does not, and how I established the difference rather than assuming it.

Several of the answers are the opposite of what I expected. Those are the parts I will spend time on.

Next: The assignment, and the problem underneath it

CORE · bullets · ~56 s spoken (121 words)

ON THE SLIDE

- A skateboard manufacturer wants **customer-designed decks**, without a designer in the loop for every order
- The assignment: a **self-trained** generator, **text + reference image**, an **interactive 3D deck**, deployed or demonstrably reproducible
- The hard part is not the pipeline. It is that **all of it has to run on one consumer laptop GPU**
- Assessed deliverable: the **research process**, not the artefact

docs/00-project-brief.md · thirteen mandatory requirements · learning outcomes D1–D7

SAY

A skateboard manufacturer wants customers to design their own decks. Today that needs a designer for every order, which does not scale to one-offs.

The assignment sets four things. The model must be self-trained, not an API call. It must take both a prompt and a reference image. The result must land on an interactive 3D deck. And it must be deployed, or reproducible by someone else.

The hard part is none of those individually — it is that all of it runs on one consumer laptop GPU.

The last line I want to plant early. This is assessed on the research process. So when I show you a failure later, I am not apologising; I am showing you the deliverable.

Next: The research question

CORE · statement · ~51 s spoken (110 words)

ON THE SLIDE

How can a **locally fine-tuned** diffusion model, conditioned on **both a text prompt and a reference image**, generate skateboard-decal artwork in **multiple visually distinct styles** with **reproducible quality** on consumer hardware with **8 GB of VRAM**?

Twelve secondary research questions, RQ1–RQ12 · report §5

SAY

I am reading this out because every clause was chosen to be falsifiable.

Locally fine-tuned — no API, and training fits on the same card. Both a prompt and a reference image — two signals that can disagree, which turned out to matter. Multiple visually distinct styles — distinct is a claim needing evidence, not an adjective. Reproducible quality, and I will come back to that phrase, because it is where my answer splits in half.

And eight gigabytes. Not "a laptop", not "modest hardware". A number, so a configuration either fits or it does not.

Underneath sit twelve secondary questions: eight answered within scope, four bounded, one component inconclusive.

Next: One constraint governs everything

CORE · statement · ~58 s spoken (125 words)

ON THE SLIDE

8187.5 MiB of VRAM. One RTX 4060 Laptop GPU.

The shipped stack peaks at **5143.73 MiB** and leaves **200 MiB spare** under real serving — **2.4 % of the card.**

EXP-019b, EXP-032, EXP-034 · experiments/registry.csv · risk R12

SAY

The number the whole project is built around.

Eight gigabytes — the measured figure the driver reports, not the marketing one. The production stack, with the base model, one adapter and the reference encoder all resident, peaks just over five gigabytes and leaves about two hundred megabytes spare while actually serving.

Two point four per cent. It fits, and it does not fit easily. I never call it headroom, because it is not.

Two consequences, both deliberate. The service runs exactly one worker — a second pipeline does not fit, so a concurrent request gets a clean refusal instead of a crash. And anything added later has two hundred megabytes to fit into.

Hold on to this number. It is behind the next three decisions.

Next: How the work was done

CORE · bullets · ~64 s spoken (138 words)

ON THE SLIDE

- Every significant choice ran a **13-step loop**: question → alternatives → criteria → experiment → **measured** result → decision → validation
- **40 registered experiments**, each with its configuration, hardware readings and conclusion
- **16 decision records** — including the ones that rejected the option I expected to pick
- **Two human review gates** with a pre-declared 1–5 rubric, fixed seeds and blinding at the first gate
- Failed experiments stay in the write-up. **A deleted failure is a deleted result.**

experiments/registry.csv · docs/decisions/ · docs/05-experiment-methodology.md

SAY

How the work was done, because the method is what is assessed.

Every significant choice went through the same loop: state the question, list the realistic alternatives, fix the criteria before running anything, build the smallest experiment that separates them, and record what happened rather than what I hoped.

That produced forty experiments and fifteen decision records. Several of those records reject the option I expected to pick — DR-007 and DR-008 are both like that, and they are next.

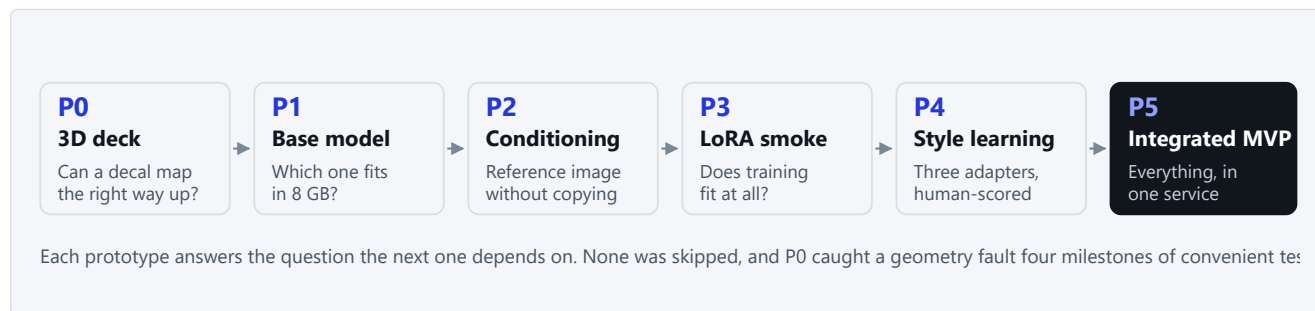
The rubric was written before the images existed, with fixed seeds and blinding at the first gate, so I could not score toward the answer I wanted. It has real limits and I will come to them.

The last point I would defend hardest: failures are in the report, not softened. Two of my strongest findings are failures.

Next: The prototype ladder

CORE · figure · ~63 s spoken (136 words)

ON THE SLIDE



SAY

Six prototypes in sequence. Each answers a question the next depends on, so I could never get several decisions deep in a direction a cheap test would have ruled out.

P0 is the one worth defending. Building the 3D deck before I had any generated artwork looks like the wrong order. It was not: it made the deck's aspect ratio and UV layout hard inputs to the image problem, and it caught an orientation fault that conveniently square test images would have hidden until the end.

P1 chose the base model, P2 how the reference gets in, P3 asked only whether training fits at all, P4 is the style learning, P5 puts it behind one service.

One thing to take from this: the awkward step early is cheaper than the discovery late. That happened three times.

Next: The dataset was built, not downloaded

CORE · split · ~54 s spoken (118 words)

ON THE SLIDE

- **148 items**, three styles, built for this project
- Ukiyo-e **55** · minimal-geometric **52** · retro-poster **41**
- Split **124 / 17 / 7** — the holdout was never seen in training
- Captions are **style-only**, not verbatim descriptions. That comparison was run, and style-only won

SAY

The dataset is mine, and it is where a lot of the work went.

A hundred and forty-eight items across three styles: ukiyo-e woodblock prints from the Metropolitan Museum, CC0; minimal-geometric, which I generated and curated; and retro silkscreen posters from the Library of Congress, public domain.

The seven-item holdout never entered a training run, so I had something honest to test memorisation against.

The captions were a real experiment, not a preference. Style-only against verbatim descriptions, on the same images and seeds. Style-only gave better prompt adherence; verbatim captions taught the model to reproduce the caption rather than the style.

If you ask about dataset size, my answer is honest and not flattering. That is slide sixteen.

Next: Licensing, decided before training

CORE · bullets · ~60 s spoken (131 words)

ON THE SLIDE

- Every item is **CC0, public domain, or created for this project**. There is no third-party licensed work in the training set
- **Licence and source URL recorded per item, in the manifest, before the item entered a split** — not audited afterwards
- Raw images are **git-ignored**; the manifest is committed. The evidence is the record, not the pixels
- Memorisation was **tested, not assumed** — nearest-neighbour checks against the holdout
- **What I cannot claim:** those checks bound copying, they do not prove it never happens

data/manifests/dataset-v1.csv · DR-006 · EXP-035 · report §10, §17

SAY

Licensing was decided before training rather than justified afterwards, and the ordering is the point.

Everything is CC0, public domain, or made by me. No third-party licensed artwork at all — so the fair-use argument never has to be made.

Each item's licence and source URL went into the manifest as a condition of entering a split. Provenance is a gate the data passed through, not a report I wrote at the end.

The raw images are git-ignored and the manifest is committed. The record is the evidence, not the pixels.

On memorisation: nearest-neighbour checks against the holdout bound the risk. They do not eliminate it, and the report says that rather than rounding up to "the model does not copy". That distinction is what I would defend if you push.

Next: Base model: SDXL was better, and was rejected

CORE · split · ~65 s spoken (141 words)

ON THE SLIDE

- SDXL scored **better** than SD 1.5 on my own rubric, at its native resolution
- It was **rejected anyway**: 10 738 MiB allocated at 1024×1024, on a card with 8187.5 MiB
- Windows **spilled into host memory instead of failing**. All thirty runs reported success
- SD 1.5 selected — not because it wins on quality, but because **it is the one that fits at a useful resolution**

SAY

The decision I expected to go the other way, and the one that set the method for everything after.

I benchmarked SD 1.5 against SDXL on identical prompts and seeds. At its designed resolution — these images — SDXL is better. My own rubric scores say so, and I left that in the report rather than quietly dropping the comparison.

I rejected it anyway: ten point seven gigabytes allocated at 1024 square, on an eight gigabyte card.

The important part is that it did not crash. Windows spills GPU memory into system RAM, so all thirty runs completed and reported success. Judged by exit code, SDXL was fine — and I would have built the whole project on it and found out in the last month.

SD 2.1 was behind an authentication wall, so this rests on two candidates, not three.

Next: The habit that came out of it

SUPPORTING · statement · ~63 s spoken (137 words)

ON THE SLIDE

A successful run is not a run that fitted.

Read every memory figure **against the ceiling**, never against whether the process survived.

EXP-005 · the measurement-methodology correction · report §19.3

SAY

Fifteen seconds on the habit that came out of that, because it is the most transferable thing I learned.

A successful run is not a run that fitted. Those are different claims, and modern systems blur them: the driver spills, the allocator caches, the OS swaps, and the process exits zero throughout.

So from that experiment on, every memory number is read against the eight gigabyte ceiling rather than against whether anything crashed. That is why I can tell you the margin is two hundred megabytes instead of telling you it works on my machine.

It also cost me one measurement error — the caching allocator holds its pool across a reset, which contaminated an experiment until I switched to one configuration per process. Documented rather than quietly fixed, because the wrong method produced perfectly plausible numbers.

Next: Reference conditioning: IP-Adapter over img2img

CORE · split · ~58 s spoken (125 words)

ON THE SLIDE

- Two ways to use a reference image: **img2img** or **IP-Adapter**
- img2img costs **zero extra VRAM** — decisive on this budget, and it lost anyway
- At the deck format it **returned its own input**. Every near-copy flag in that milestone came from img2img
- **IP-Adapter selected**, scale 0.55. *A method that reproduces its input is not a generator.*

SAY

The decision where the criteria mattered most.

Two realistic ways to condition on a reference. img2img starts the diffusion from the reference itself. IP-Adapter encodes it and injects it through cross-attention, which costs extra VRAM.

On a card with two hundred megabytes spare, "costs extra VRAM" should have ended the argument. img2img is free.

Look at the pairs. Left is the reference the user supplied, right is what img2img returned at the deck format. Several are perceptually the same image. Every near-copy flag in that milestone came from img2img; none came from IP-Adapter.

So I rejected the cheaper method on what it does, not what it costs. A system that hands back the customer's own upload with a filter on it is not a generator.

Next: Why LoRA, and what that does not claim

CORE · bullets · ~61 s spoken (132 words)

ON THE SLIDE

- Full fine-tuning does not fit. Training from scratch needs **millions of images and a cluster** — neither exists here
- LoRA trains a **small pair of low-rank matrices** beside the frozen base model
- Measured: **rank 8**, 3 133 MiB at 512², 300 steps in **91 seconds**. It fits with room to work
- Each adapter is **6414480 bytes** — the styles ship as megabytes, not gigabytes
- **The claim is feasibility, not superiority.** Four alternatives were screened on criteria and never run

DR-009 · EXP-016 to EXP-019b · RQ1, recorded as bounded

SAY

Why LoRA, and — just as important — what that does not claim.

From scratch was never realistic: a base diffusion model takes hundreds of millions of images and a cluster. I have one laptop and a hundred and forty-eight pictures. Full fine-tuning updates every weight, and the optimiser state alone does not fit in eight gigabytes.

LoRA freezes the base model and trains a small pair of low-rank matrices beside it. Measured here: rank eight, about three gigabytes during training, three hundred steps in ninety-one seconds. It fits with room to iterate.

The honest bound, before you ask: I did not prove LoRA beats the alternatives, only that it is feasible here. Textual Inversion, DreamBooth and full fine-tuning were screened on criteria and never run, so RQ1 is bounded, not answered.

Next: Training longer made it less obedient

CORE · two-figures · ~67 s spoken (146 words)

ON THE SLIDE

Training longer made the model more stylish and less obedient. Two of three styles ship at step 300. *Ukiyo-e* is the counterexample — it kept improving and ships at step 600.

SAY

My favourite result, and I only have it because of a process decision.

Same style, same prompts, same seeds. Left is three hundred training steps, right is six hundred.

Six hundred is more stylised — and less obedient. Prompt adherence dropped from four to three while style consistency held at five. The model got better at the style by getting worse at listening to you, and on a product where the customer types what they want, that is the wrong trade.

So two of three adapters ship from step three hundred, not the six hundred they trained to. *Ukiyo-e* is the exception and ships at six hundred. That is why the checkpoint is chosen per style.

I only saw it because I checkpointed four times along each run and had them scored blind. One global step count would have got two styles out of three wrong.

Next: What actually ships

SUPPORTING · table · ~56 s spoken (121 words)

ON THE SLIDE

style	experiment	ships at	verdict
minimal-geometric	EXP-027	step 300	PASS
ukiyo-e	EXP-028	step 600	PASS
retro-poster	EXP-029	step 300	PARTIAL PASS

Applied at weight **0.7**. Every checkpoint is **verified by SHA-256 on every request** — the weights cannot be rebuilt, so they are treated as artifacts, not recipes.

docs/deployment/weights-manifest.md · guarded by pytest so the manifest cannot drift from the code

SAY

What actually ships — a table rather than a claim, so you can check it.

Three adapters, three different checkpoint steps, three separate experiments. Two are clean passes against the rubric. One is a partial pass, and that is the next slide.

Default weight zero point seven, which is where the blind scoring landed. It stays adjustable in the interface.

The last line matters more than it looks. Each checkpoint's hash is verified when the style is activated, on every request — not once at startup. These weights cannot be regenerated, so the files are the authority, and an integrity check on every request is what catches a swap or a corruption.

Why they cannot be regenerated is the determinism slide.

Next: One style is a partial pass

CORE · split · ~64 s spoken (139 words)

ON THE SLIDE

retro-poster ships as a PARTIAL PASS.

- It learned the **frames, borders and display typography** of the archive posters, not only their style
- Shipped **with the limitation stated in the interface**, on every request
- Not dropped, not quietly upgraded, not re-scored until it passed

SAY

One of my three styles is a partial pass, and I would rather present it than have you find it.

The training material is WPA silkscreen posters from the Library of Congress — so almost all of it has a frame, a border and large display lettering as part of the composition.

The adapter learned all of that. Ask for a mountain and you often get a mountain inside a poster frame with lettering that is not quite text. Not what a decal needs.

Three options: drop the style and lose a third of the deliverable, re-score it until it passed, or ship it labelled. I shipped it labelled — marked partial in the interface, so the user is told before choosing it.

I know the likely fix, and it landed after the freeze. Future work, not a claim.

Next: How many images does a style need?

CORE · bullets · ~68 s spoken (148 words)

ON THE SLIDE

How many images does a style need? I do not know. **INCONCLUSIVE**

- Trained the same style on **12, 24 and 44 images**, everything else held fixed
- The ordering came out **non-monotonic** — more images was not reliably better
- **No minimum count was established**, and none is claimed
- Reported as inconclusive rather than rounded into a recommendation

EXP-024n12, EXP-024n24, EXP-024 · RQ4, image-count component · report §13

SAY

A question I set out to answer and did not.

RQ4 has two halves. The caption half worked: style-only captions beat verbatim ones, and that result is in production.

The image-count half did not. Same style on twelve, twenty-four and forty-four images, everything else fixed. I expected a curve with a plateau I could point at. What I got was non-monotonic — the middle condition did not sit where it should have. With one run per condition I cannot separate a real effect from variance, and repeating them enough times did not fit in the GPU budget.

So it is inconclusive, and no minimum image count is claimed anywhere in the report.

Why this slide exists: it would have been easy to write "about forty images is enough". It is roughly what I used, it sounds reasonable, and nobody would have checked. I just did not measure it.

Next: The rest of what did not work

SUPPORTING · bullets · ~74 s spoken (160 words)

ON THE SLIDE

- **SD 2.1 could never be measured** — authentication-gated. The model decision rests on two candidates, not three
- **Memorisation is bounded, not settled.** The holdout checks limit copying; they do not disprove it
- **Style conditioning can dominate a detailed prompt** — ask for a skyline *and* a skateboarder and you may get neither, beautifully
- **Evaluation:** one human approver, **AI-assisted visual analysis**, and **no second independent human rater** — so no inter-rater agreement can be reported
- Institutional sources returned **HTTP 429 and 403** across five official paths; the wording that depended on them was removed

report §12 failed and blocked experiments · §18.4 validity threats

SAY

Quickly — a limitations slide that is only a formality is worse than none.

SD 2.1 sat behind an authentication wall, so the base-model comparison has two candidates, not three. A gap in the evidence, not a verdict on SD 2.1.

Memorisation: the holdout checks bound the risk. They do not settle it.

Third is the limitation a user actually hits. Style conditioning can overwhelm a detailed prompt — ask for a skyline with a skateboarder and you can get a beautifully consistent deck with neither clearly in it. Measured, documented, not fixed.

Fourth, and I would rather raise it than have you find it. Scoring was AI-assisted: ChatGPT proposed visual analysis and scores at the gates, I reviewed and approved every recorded score and kept final authority. But there was no second independent human rater, so I cannot report inter-rater agreement. An earlier draft overstated that separation; I caught it at a gate and corrected it in nine places.

Next: The integrated system

CORE · figure · ~62 s spoken (135 words)

ON THE SLIDE

Browser

React 19 + Vite
Three.js / R3F deck
Procedural geometry,
runtime texture swap

```
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<text x="428" y="116" font-size="21" fill="#46525f">uvicorn, <tspan font-weight="700" fill="#11151c">one worker</tspan></text>
<text x="428" y="148" font-size="21" fill="#46525f">Busy lock &#8594; 409, never OOM</text>
<text x="428" y="180" font-size="21" fill="#46525f">Upload validation, timeouts</text>
<text x="428" y="206" font-size="21" fill="#46525f">SHA-256 checkpoint verify</text>

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<text x="858" y="116" font-size="21" fill="#c8d0dc">Stable Diffusion 1.5</text>
<text x="858" y="148" font-size="21" fill="#c8d0dc">+ one per-style LoRA</text>
<text x="858" y="180" font-size="21" fill="#c8d0dc">+ IP-Adapter (optional)</text>
<text x="858" y="206" font-size="21" fill="#8fa0ff">512 &#215; 1536, direct</text>

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<text x="1306" y="186" font-size="24" font-weight="700" fill="ffffff">8 GB</text>

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<text x="342" y="126" font-size="18" fill="#7b8794">HTTP</text>

<text x="4" y="296" font-size="22" fill="#46525f">Two processes, one machine. Single-worker is a correctness requirement enforced in
```

SAY

Briefly, because the interesting part is a constraint rather than a diagram.

Two processes on one machine: a React frontend with the Three.js deck, and a FastAPI service that owns the model.

The single worker is what I would point at. It is not a default I never changed — it is asserted in code at startup, because a second resident pipeline does not fit. Two simultaneous requests, and the second gets a clean refusal rather than an out-of-memory crash that takes the first down with it. Refusing correctly is a feature here.

Uploads are validated properly — allowlist, real decode, size limits, random internal filenames, no user string reaching a path. An upload endpoint attached to a GPU is worth attacking.

The deck geometry is procedural, so no third-party model licence enters the deliverable.

Next: The MVP, on screen

CORE · split · ~72 s spoken (156 words)

ON THE SLIDE

- Generation is **direct to the deck format**, 512×1536 — not square-then-crop
- The progress bar reports **real diffusion steps**, from the pipeline
- While the model loads it shows **no percentage at all**, because loading exposes none. *A bar creeping to 90 % would look better and be a lie*
- Every result carries its **seed, pinned model revision, adapter hash and measured VRAM**

SAY

Two details I would defend.

First, generation goes straight to the deck's tall aspect ratio rather than generating a square and cropping. I expected that to hurt — diffusion models are trained mostly on square images and drift at extreme ratios. I wrote it down as a hypothesis, tested it, and it was refuted. That is RQ8, and it saved me a workaround I did not need.

Second, the progress bar shows real denoising steps from the pipeline — step fourteen of thirty is actually fourteen of thirty. And while the model loads it shows no percentage at all, because loading exposes none. I could have drawn a bar creeping to ninety per cent. It would look better and it would be a lie about where the time went.

Every image carries its seed, the pinned model revision, the adapter's hash and the measured VRAM. It is traceable to a specific experiment, not just a picture.

Next: A trade-off the interface admits to

SUPPORTING · two-figures · ~64 s spoken (138 words)

ON THE SLIDE

Decals are 1:3. The deck surface is 1:3.9. **Both options were built and compared.** The interface tells the user which one it chose, and that the artwork is stretched.

SAY

A small thing that shows how the decisions were made.

Decals are one to three. The deck surface is one to three point nine. Something has to give.

I built both options rather than picking one. Stretch to fill, which distorts the artwork about thirty per cent vertically. Or preserve the ratio, which leaves a quarter of the board bare.

I chose to fill, because a graphic that covers the deck reads as a finished product and one with a bare strip reads as a bug.

The part I would defend is the last line: the interface says so. It tells the user the artwork is stretched, in the panel, rather than hoping nobody measures it. A trade-off the user is told about is a design decision. The same one hidden is a defect waiting to be found.

Next: Testing

CORE · bullets · ~73 s spoken (158 words)

ON THE SLIDE

- **473** product and research tests + **16** report-validation tests = **489** **pytest**
- **183** frontend unit tests · **38** browser end-to-end scenarios
- The **clean-clone test** is the one that counts: clone into an empty directory, install from scratch, reach a working system
- It caught an **integrity control that had only ever passed on the machine that wrote it**
- A green suite does not prove the generator makes good pictures. **Only the human gates did that**

[docs/evidence/M8/](#) · [docs/evidence/M8/clean-clone/](#) · [report §15](#)

SAY

Testing, and one result that changed how I think about testing.

I keep the pytest number split on purpose. Four hundred and seventy-three tests cover the product and the research code. Sixteen more test the report itself — that every quantitative claim still matches the evidence file it came from. Four hundred and eighty-nine altogether, plus the frontend and browser suites.

The one that earned its place is the clean-clone test: clone into an empty directory, install from scratch, reach a working system, about ten minutes.

It caught a real defect. A checkpoint integrity control passed every time locally — because the path it checked only existed on my machine. In a clean clone it failed immediately. It had been green for weeks while verifying nothing.

The honest limit: those tests tell you the code does what I told it to. Not one of them tells you a generated deck looks good. Only the human gates did that.

Next: Determinism has halves

CORE · split · ~70 s spoken (152 words)

ON THE SLIDE

- **Inference is deterministic.** A clean clone reproduced an earlier output **byte for byte**, three days later, in a freshly built environment
- **Training is not.** The adapter initialisation draws from an **unseeded global generator**, so a run cannot be repeated from its recorded seed
- So the checkpoints are **artifacts, verified by hash** — not recipes
- A defect that produces perfectly valid-looking runs, and is invisible unless you compare weights

SAY

The primary question asks for reproducible quality, and this is where my answer splits in half.

Inference is deterministic — and not just on my machine on the same day. A clean clone in a fresh environment, three days later, reproduced an earlier output byte for byte. Same SHA-256.

Training is not reproducible at all from its recorded seed. Two runs with identical configuration and an identical seed produced different weights, because the adapter initialisation draws from an unseeded global generator. The seed I was diligently recording never governed the part that mattered.

So the checkpoints are artifacts, not recipes: verified by hash on every request, and the deployment docs say plainly that three required files cannot be regenerated.

The part worth pushing on: that defect produces runs that look completely fine. It is invisible unless you compare weights — which I only did because I tried to reproduce my own result.

Next: Deployment: local, and chosen

CORE · bullets · ~69 s spoken (150 words)

ON THE SLIDE

- Four options were compared. **Selected: native local run + pre-generated backup demo assets** (Option D of four)
- **200 MiB spare prevents a second resident pipeline on the validated GPU**, so multi-worker inference is unsupported
- Docker with GPU passthrough was **never verified on this machine**; its VRAM overhead is unmeasured
- **Public cloud deployment was deliberately not selected for this bachelor version** — it is a cost and a network dependency on demo day, for no research gain
- Reproducibility is carried by the **runbook and the clean-clone proof**, not by a public URL

DR-014 · alternatives A–D compared on nine criteria · EXP-034

SAY

"Is it actually deployed" is a fair question and the answer needs no spin.

There is no public URL. It runs locally, and that was a choice between four compared options rather than something I ran out of time for.

Not cloud, because renting a GPU costs money for the length of the project, adds a network dependency on the one day everything must work, and — the deciding part — answers nothing. The question is what fits in eight gigabytes of consumer hardware. A rented A100 replaces that question rather than answering it.

Not Docker, because GPU passthrough was never verified here and its overhead is unmeasured. On a two hundred megabyte margin, unmeasured overhead is disqualifying.

What I do claim is reproducibility, and it rests on the runbook and the clean-clone test — a stronger claim than a URL, which only proves it runs where I put it.

Next: Conclusions

CORE · bullets · ~66 s spoken (144 words)

ON THE SLIDE

Yes — and the configuration that fits is now specific and measured. SD 1.5 at 512×1536, one per-style LoRA at 0.7, IP-Adapter at 0.55, one resident pipeline.

Bounded three ways: **the margin is 2.4 % · reproducible describes inference, not training · "multiple distinct styles" is three, and one is partial.**

Not concluded: that LoRA beats the alternatives · that SD 1.5 is the best base model · that three styles is enough · that any image count is a threshold · that a green suite means good pictures.

report \$19 · eight RQs answered in scope, four bounded, one component inconclusive

SAY

The answer is yes, and the useful part is the specificity: I can name the configuration that fits, with the memory reading behind each element.

But the bounds matter more than the yes, so I state all three every time. The margin is two point four per cent — it fits, not comfortably. Reproducible describes inference, not training. And multiple distinct styles means three, one of which is partial.

The second half is what I would most like to be judged on: things I could easily have concluded and did not, because I did not measure them. That LoRA beats the alternatives. That SD 1.5 is the best base model. Any minimum image count. That a green test suite means good pictures.

Each of those is one easy sentence. None is supported by what I measured, so none of them is in the report.

Next: Live demo

CORE · demo · ~114 s spoken (246 words)

ON THE SLIDE

Prompt → style → **512×1536 decal** → 3D deck. Running on this laptop's GPU. Nothing leaves the machine.

If the GPU does not cooperate, that is a live-demo problem and not a project problem — there is a recorded run ready and I will say which one you are looking at.

SAY

Hand off to the timed demo script here. Four minutes. Do not improvise it.

Opening line, then move to the browser — it should already be at the printed URL, one tab, no console, notifications off:

"This runs locally, on this laptop's GPU. The three styles are not prompt tricks — they are adapters I trained on the dataset I showed you."

Press Generate at 0:45 and keep talking. A warm generation is about thirteen seconds, a cold one about thirty. Both are dead air if you stop, and both are enough time to explain the progress panel if you do not. Suggested prompt: *a mountain and a rising sun*, style **Ukiyo-e** — the unambiguous pass, and it reads well at a distance.

Do not generate with retro-poster. It is a partial pass; showcase it in words and generate with a full pass.

Then: reproducibility metadata, the 3D deck with the stretch disclosure, and upload-your-own-decal at 2:30 — which is also the fallback, so the room has already seen it work.

If it fails: say the line on the slide, then go to `Upload your own decal` with a pre-generated PNG. Do not reload, do not retry, do not open a terminal, do not run a second generation. The fallback ladder has four rungs and every one of them is under thirty seconds.

If the output ignores half the prompt — **that is not a failure, it is slide seventeen**. Say so and use it.

Hand off to the demo here. The timed script and the four-rung fallback ladder are separate documents — do not improvise either.

CORE · bullets · ~73 s spoken (159 words)

ON THE SLIDE

- **D1 / D5 / D6** — 40 experiments, six prototypes, conclusions bounded by what was measured
- **D3 / D2** — 16 decision records, a planning log that records what changed and why, public issues and board
- **D4** — alternatives compared and rejected on criteria: base model, conditioning, fine-tuning, deployment
- **D7** — a 91-page report, this deck, and a system that states its own limitations to its user
- **What I would change:** repeat every condition before drawing a curve · seed *everything* and verify by comparing weights, not by trusting the log · get a second rater

[github.com/KylianAlgoet/Selftrained-and-deployed-AI-image-generator · docs/learning-outcome-traceability.md](https://github.com/KylianAlgoet/Selftrained-and-deployed-AI-image-generator/blob/master/docs/learning-outcome-traceability.md)

SAY

Briefly against the learning outcomes, then what I would change.

The research outcomes are carried by the experiment registry, the prototype ladder, and conclusions that stop where the evidence stops. The professional ones by the decision records and a planning log that records what slipped and why, rather than being rewritten to match the original plan.

Two things I would change, and one is embarrassing.

Repeat every condition before drawing a curve. The image-count question is inconclusive because I ran each condition once — a planning mistake, not a resource one.

And seed everything, then verify by comparing weights rather than trusting the log. I recorded a seed for every run and was confident about reproducibility for weeks. It was not governing the initialisation. I found out only because I tried to reproduce a result instead of assuming I could — which is, in the end, what this project taught me.

Thank you. I am happy to take questions.

Next: — end of deck —